PeptidoformVisualisation: a Shiny app to visualise the intensity patterns of different peptidoforms

BACKGROUND & METHODS

The lack of open source, freely available visualisation and preprocessing tools to visualise (histone-related) peptidoforms is a missed opportunity to make the most out of histone post translational modification (hPTM) data.

Drugs that target epigenetic mechanisms like histone posttranslational modifications (hPTMs), called epidrugs, can be used in clinical trials trying to reprogram the epigenetic landscape in the cancer epigenome. This holds great promise in patient care to prevent disease states. Nonetheless, the mapping of hPTMs, with the aid of mass spectrometry based proteomics (MS) has not been extensively described. Therefore, we have made the first untargeted map of the histone code for 21 T-cell acute lymphoblastic leukaemia cell lines. However, extracting biological information from this map is not trivial because of the complexity of the histone code[1-2]. Because of this, data sharing and reuse is important to facilitate transfer of knowledge and information. We have pre-processed the raw data into a single platform-independent Progenesis QIP project that is fully accessible so that other researchers can use this browsable template. Because data visualisation and pre-processing are important first steps when analysing data but are often not trivial for biologists without programming experience, we have developed an accompanying Shiny app in which users can intuitively visualise the different peptidoforms present in the dataset.
This Shiny app, called PeptidoformVisualisation, is an interactive web app developed with the Shiny R Package (https://shiny.rstudio.com/, version 1.7.1). The app and installation instructions can be found via the link on the right.

WHAT'S NEW?

We developed a Shiny app that pre-processes and visualises peptidoform-level intensity data. This app has been made to accompany the largest histone atlas to date for T-ALL cells, but any peptidoform data can be uploaded.

WHERE TO FIND?

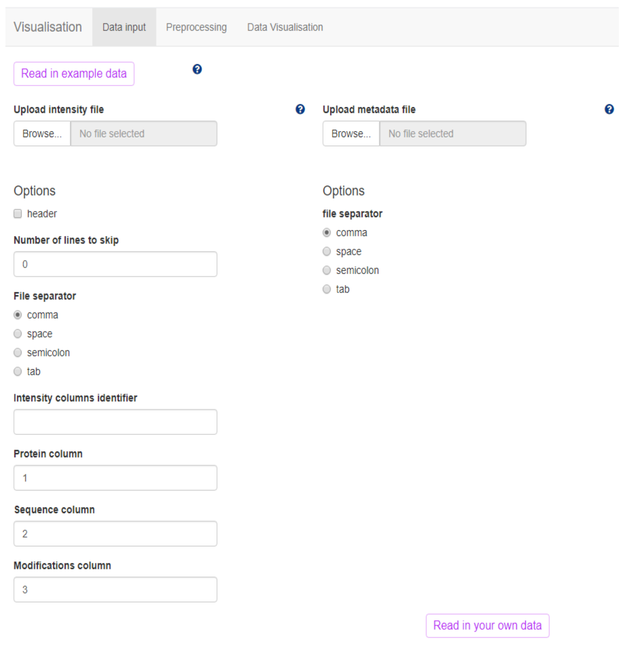
The data: http://proteomecentral.proteomexchange.org/cgi/GetDataset?ID=PXD031500

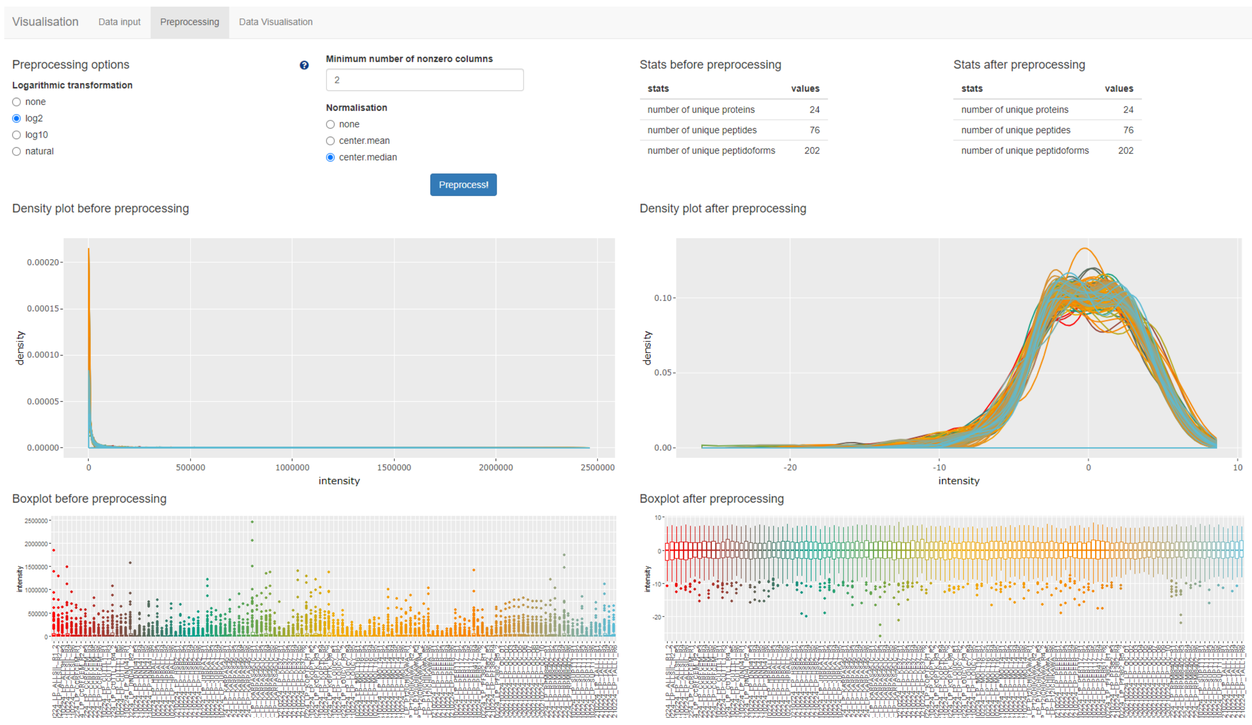
The app: https://github.com/statOmics/PeptidoformViz

Paper: https://doi.org/10.1101/2022.05.05.490796

THE APP – tab 1 & 2

The app currently consists of three main panels: data import, pre-processing and visualization. There are help buttons everywhere to assist the user. The data input tab requires a file that contains peptide level intensity data and a corresponding metadata file containing at least enough information so that each row represents one unique ID. In the pre-processing tab the user can opt for several pre-processing steps: logarithmic transformation, normalization and a filtering step based on missing intensity values. A density plot and boxplot of the data are generated to aid the user in this decision.





THE APP – tab 3

After the pre-processing, the user can go on to the visualization panel. Here we work with the peptidoforms of one protein only. Accordingly, the user can select a protein of interest present in the dataset via a dropdown menu. A data table is displayed containing all peptidoforms of the selected protein and corresponding intensity values. It is possible to search for certain patterns in peptide sequences in the table, e.g., a present modification or a certain sequence. Two figures are rendered based on the data in the displayed data table: a boxplot and a line plot of all the peptidoforms. It is possible to highlight peptidoforms of interest by clicking on them in the data table. The corresponding line(s) in the lineplot will then be highlighted, and corresponding dots will appear in the boxplot. The data table and figures can be downloaded to the user's computer.

WHAT'S NEXT?

Because of the nature of Shiny apps and how they are developed, it is quite easy to add additional features later on. In the future, we aim to extend the functionalities of the Shiny app with advanced statistical analyses, including single PTM level quantification.

GET IN TOUCH

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References

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